27 RS232 interface features

1. **General features**
   The balance transmits the value visualized on the display following serial RS232C standard, allowing to print the value of weight to a PC monitor or to a serial printer. In the case of connection to a PC, it will be possible to select the transmission in continuous mode or transmission at user command through pressing of the **PRINT** button (as described at par.0). The balance is also capable of receiving commands, always through the standard RS232C, that allow to perform all the functions available through the keyboard of PC itself. The speed of transmission and reception can be selected, as described previously (see par. 0), to 1200, 2400, 4800, e 9600 baud. The character format is of 8 bit preceded by one bit of start and followed by a bit of stop. Parity is not considered.

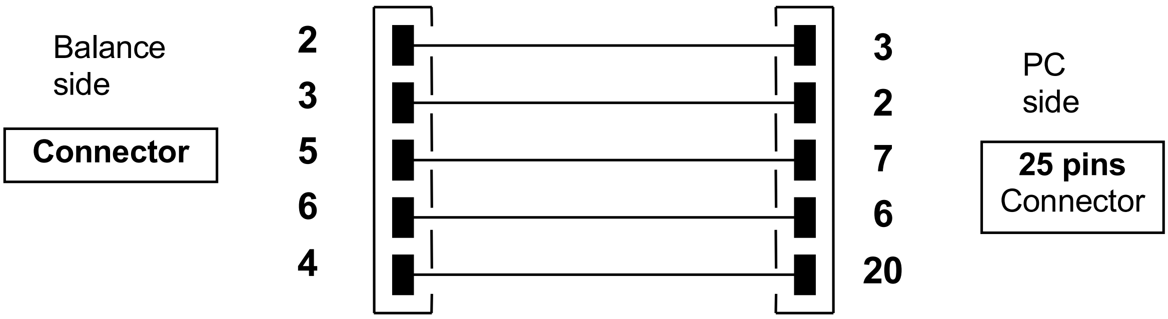
2. **Selection of interface for PC**
   Selecting the transmission to PC (personal computer IBM compatible), it is achieved a continuous transmission output, at the same rate of weight update on display of the balance. It is possible to perform all the functions of the balances directly from the computer keyboard, transmitting to the balance the ASCII codes as in the table below. The connector to use for connection to PC is the number 4 in fig.1 of par. 0

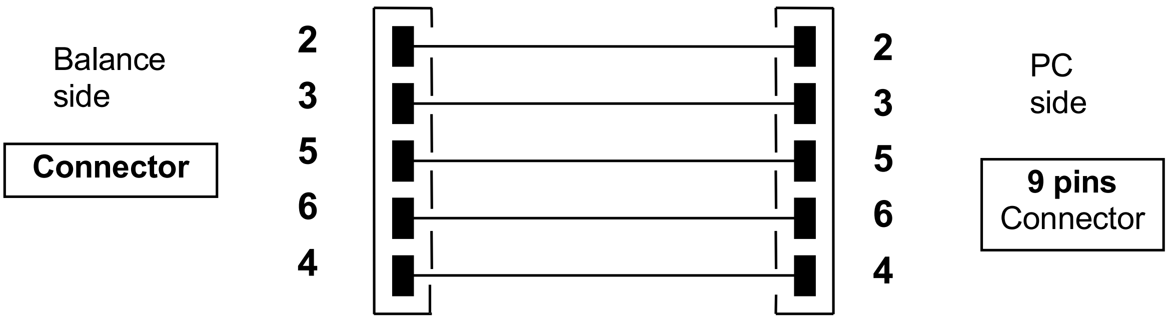
CODE	1st FUNCTION (SINGLE PRESS)
“T” = H54	TARE
“C” = H43	CALIBRATION
“E” = H45	ENTER
“M” = H4D	MENU
“O” = H4F	ON/OFF

CODICE	2nd FUNCTION (PROLONGED PRESS)
“t” = H74	TARE
“c” = H63	CALIBRATION
“e” = H65	ENTER
“m” = H6D	MENU
“o” = H6F	ON/OFF

Selecting the transmission to PC at user command, it is achieved a transmission output only when the **PRINT** button is pressed, also in this case it is possible to perform all the functions of the balance directly from the keyboard of the computer, sending to the balance the ASCII codes in the table above. The connector to use for connection to PC is the is the number 4 in fig.1 of par. 0

3. **Connection of the balance to PC**
To receive/transmit data, connect the connector (number 4 in fig.1 of par. 0) of the balance to the serial port of the PC as shown below:





4. **Transmission format**
String transmitted is composed by the following 14 characters:
- First character: weight sign (blank or -)
- second/ninth character: weight or other data
- tenth/twelfth character: weight unit symbol
- thirteenth character: stability indicator (only for continuous transmission)
- fourteenth character: carriage return
- fifteenth character: line feed

Eventually non-significative zero are spaces.

In the following tables the various transmission formats are shown:

Weighing mode (valid both for continuous transmission and transmission at user command)

1°	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°
Sign	Weight								Weight unit			Stability*	CR	LF

*only for continuous transmission

Density mode (only in transmission at user command mode)

1°	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°

d	=	Density value	Space	Weight unit	CR	LF

Piececounting mode (only in transmission at user command mode)

Number of pieces

1°.	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°
Pcs			:	spaces				Number of pieces							

Total weight of pieces

1°.	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°	18°	19°	20°
Weight						:	space	Value of weight								space	g	space	S

Average unit weight of pieces:

1°.	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°	18°
PMU			:	spaces				Value of weight								spaces	g

Percentage weight mode (only in transmission at user command mode)

Percentage

1°.	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°	18°
Perc				.	space			Percentage								space	%

Weight

1°.	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°	18°
Weight						space		Weight value								space	g

Animal weighing mode (only in transmission at user command mode

Time

1°	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°	18°

Time	space	=	space	Time value	Sec	space

Averaged Weight

1°.	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°	18°
Ave			.	=	space				Averaged weight value						space		g

Totalizing (only in transmission at user command mode)

Weighing

1°.	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°	18°
Weighing number		.	space			Value of weight								space		g	

Weighs total

1°	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°	18°
S	space	=	space			Value of weight								space		g	

Threshold function (only in transmission at user command mode)

1°	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°	18°	19°	20°
Weight						:	minus si ng if negative	Value of weight								s p a c e		g	

If Low

1°	2°	3°	4°	5°
- Low -				

Else Hight

1°	2°	3°	4°	5°
- Low -				

Else Ok

1°	2°	3°	4°	5°
- Ok -				

5. **Selection of interface to printer**

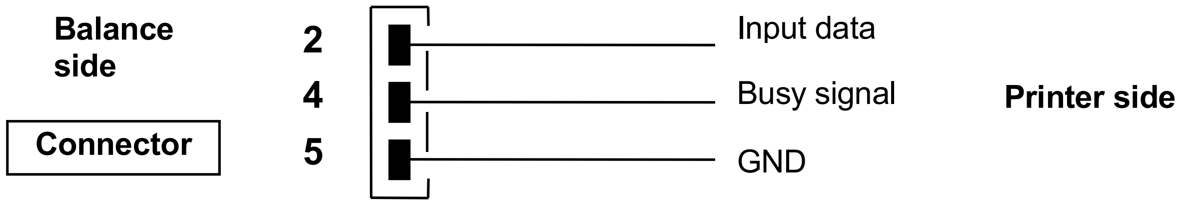
Selecting the PRINTER mode, the serial output of the balance is set to work with serial printers.

In this case the printing is effected only when the **PRINT** button is pressed and with stable weight. If the stability is not reached within ten seconds, the message **ERROR05** is displayed preceded by a short acoustic signal and the value of the weight is not sent to the printer.

The connector to be used for the connection is the number 4 shown in fig.1 of par. 0

6. **Connection of the balance with the serial printer**

Connect a serial printer to connector 1 to the balance as shown in the following scheme:



If the optional TLP50 printer is used, it will be possible to print both in *continuous-mode* and in *labels-mode* with the following formats:

Weighing mode and maximum load mode

```
12-02-2008          12:00

Weight:          22.000 g
```

Piececounting mode

```
12-02-2008          12:00

Pcs              100
Weight:   300.000 g
PMU:        3.000 g
```

Density determination mode

```
12-02-2008          12:00

d=   2.80066  g/cm3d
```

Percentage weight mode

```
12-02-2009          12:00

Perc.           100.0%
Weight:   300.000 g
```

Animal weighing mode

```
12-02-2010          12:00

Time =  6 Sec

Ave. =  59.446 g
```

Totalizing (sum of weights) mode

```
12-02-2009          12:00

1.          16.589    g
2.          17.226    g
…
99.
------------------------------
S=         33.815    g
```

12-02-2013	12:00
Weight: 0.00g -LOW-	

12-02-2012	12:00
Weight: 49.20g - OK -	

12-02-2011	12:00
Weight : 249.42g -HIGH-	

7. **Connection of the balance to the optional alphanumeric keyboard**
   To connect the optional alphanumeric keyboard it must be used the connector number 4 shown in fig.1 of par. 0 (the same used for connection to a PC). In this case the connection to PC or to printer must be effected through the connector placed on the optional alphanumeric keyboard.

8. **Map of RS232 interface (Indicated to par 0, Fig.2)**

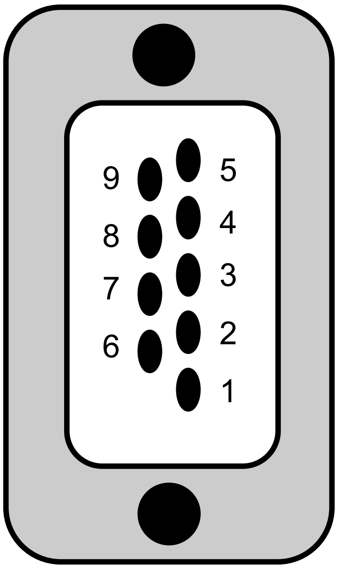


Fig. 2

CONNECTOR MAP (CONNECTION FOR KEYPAD, PC AND PRINTER)

- pin 1 = Power +5V for keyboard
- pin 2 = Tx signal
- pin 3 = Rx signal
- pin 4 = busy signal
- pin 5 = Gnd
- pin 4-6 = connected to each other for connection to PC

Fig. 3